



Service Bulletin Number: 3666052-03

Released Date: 15-mar-2007

Camshaft Reuse Guidelines for Cummins® Engines with Roller Followers or Roller Tappets

Camshaft Reuse Guidelines for Cummins® Engines with Roller Followers or Roller Tappets

General Information

This bulletin applies specific visual inspection criteria for all Cummins® engines with roller followers or roller tappets in contact with the camshaft lobe surfaces. This bulletin addresses the instances in which surface deterioration is predominately the result of galling or spalling. An additional resource for camshaft information is Technical Overview of Camshaft Durability, Bulletin 3379031. Cummins Inc. Distributors and Dealers are encouraged to use their technical expertise to implement these guidelines for low cost, high quality repair of Cummins Inc. products.

Although this bulletin applies to all Cummins® engines with roller followers or roller tappets, it was revised to include knowledge gained from an injector camshaft engineering project completed on the Signature, ISX, QSX15 Heavy Duty engine. Based on the results of engineering analysis and endurance testing, the acceptable width for lobe galling damage on the injector camshaft was increased specifically for the Signature, ISX, QSX15, and ISX with CM870 control module engines. The updated lobe galling width limit applies for the surface damage indicated in Figures E and F.

Definitions

Polishing

- Normal surface condition where machining lines have been smoothed from contact between mating parts (see Figure B).

Denting

- A depression left in the surface by a piece of foreign material that is trapped between the lobe and roller. A dent has a relatively smooth, shiny bottom and does **not** have rough or sharp edges (see Figure C).

Frosted Bands

- High-density microdenting. Frosted (white) in appearance (see Figure D).

Varnished Bands

- Surface film of oxidized oil. Brown or blue in appearance. Varnished bands are **not** detectable with a fingernail (see Figure E).

Galling

- Transfer of small pieces of material between the follower roller and the camshaft lobe surface by welding one surface to the other (see Figures F and G). Galling typically occurs when a follower roller skids on the camshaft lobe surface. This damage is detectable with a fingernail. Multiple pictures are shown below to provide examples of what galling can look like.

Pitting

- Loss of a piece of material from the lobe resulting in a hole that is visible to the naked eye. Pits typically have rough, dark bottoms and sharp edges, and usually occur in the heavily-loaded areas of the lobe (see Figure H). Multiple pictures are shown below to provide examples of what pitting can look like.

Macro-Spalling

- Loss of large pieces of material from the lobe surface (see Figure I). This amount of damage will affect the operation of the engine. Multiple pictures are shown below to provide examples of what macro-spalling can look like.

Reuse Guidelines

Analysis of lobe deterioration, as represented in the photographs in Figures A through I, can be made objective by following the reuse information presented in Table 1, Figures A through I.

Camfollower Roller Inspection

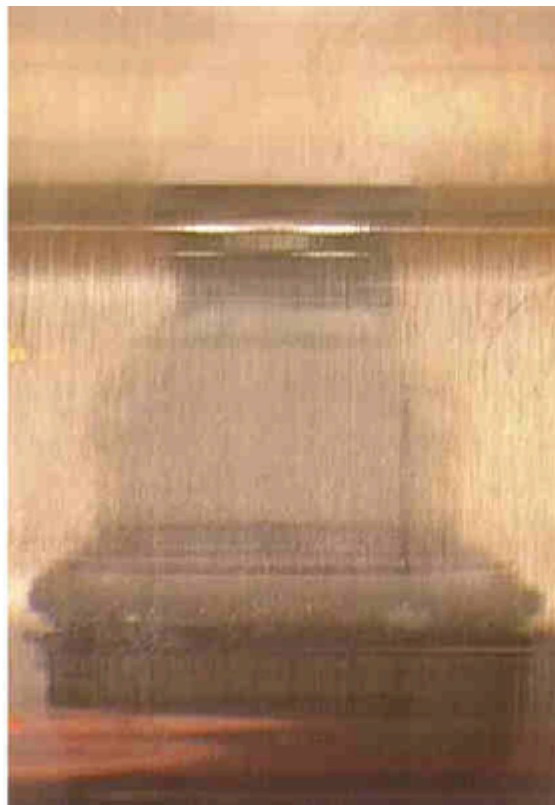
Inspect and replace as necessary all camfollower rollers and pins, or assemblies. Use the criteria from this bulletin for visual inspection of the rollers. Reference the appropriate Engine Shop or Service Manual for other reuse guidelines.

Lubricating Oil System Inspection

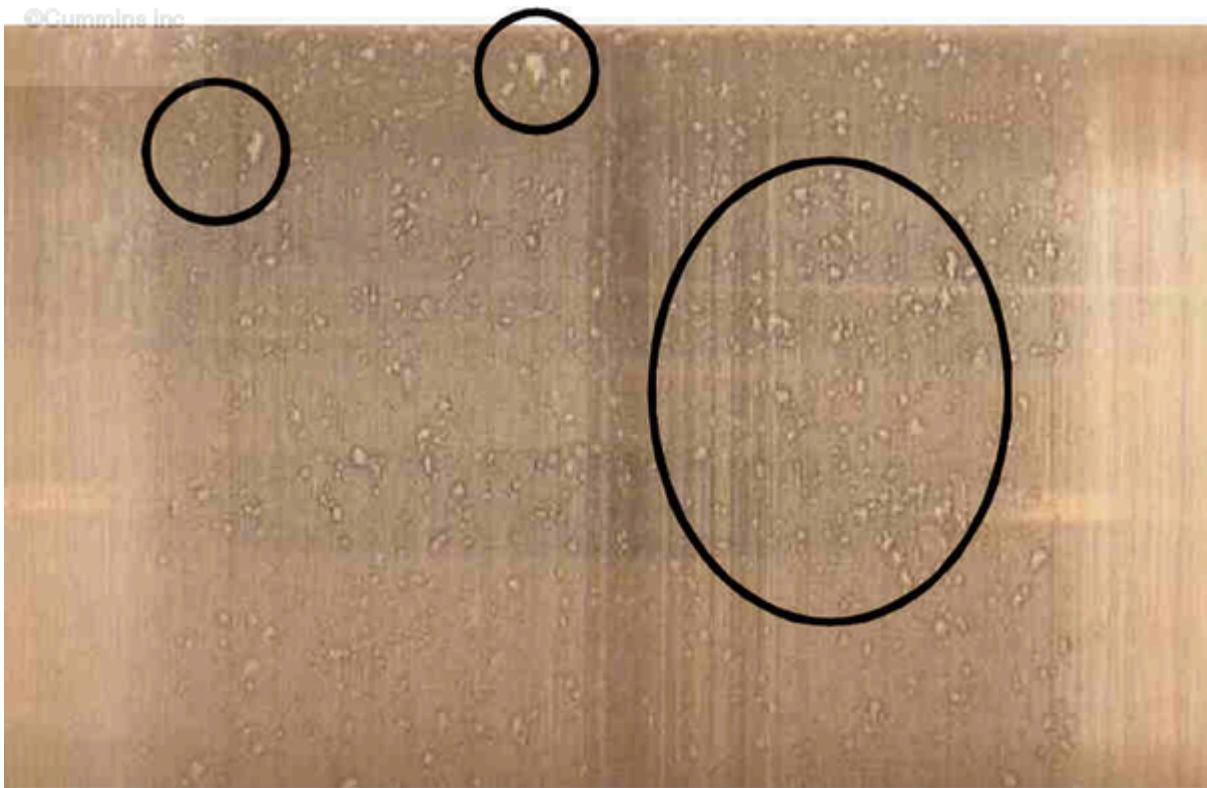
In the event that macrospalling is observed, inspect the lubricating oil filter for metallic debris. If metallic debris is found in the filter, inspect and replace as necessary the main and rod bearings, camshaft bushings, turbocharger, lubricating oil pump and pressure regulator, and the lubricating oil cooler. For further information about these components, please refer to the service procedure and the reuse guidelines for each component.



A. New Lobe

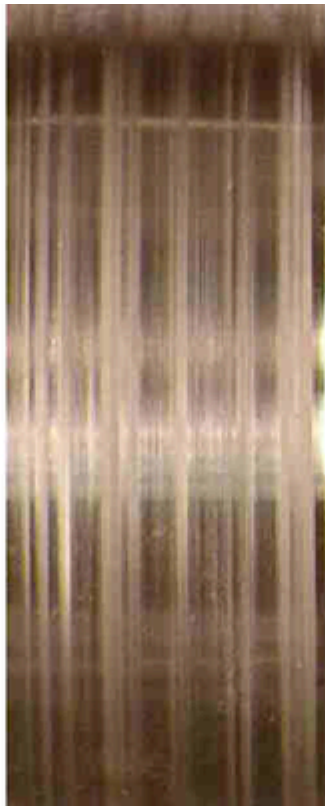


B. Polishing: Unconditional Reuse

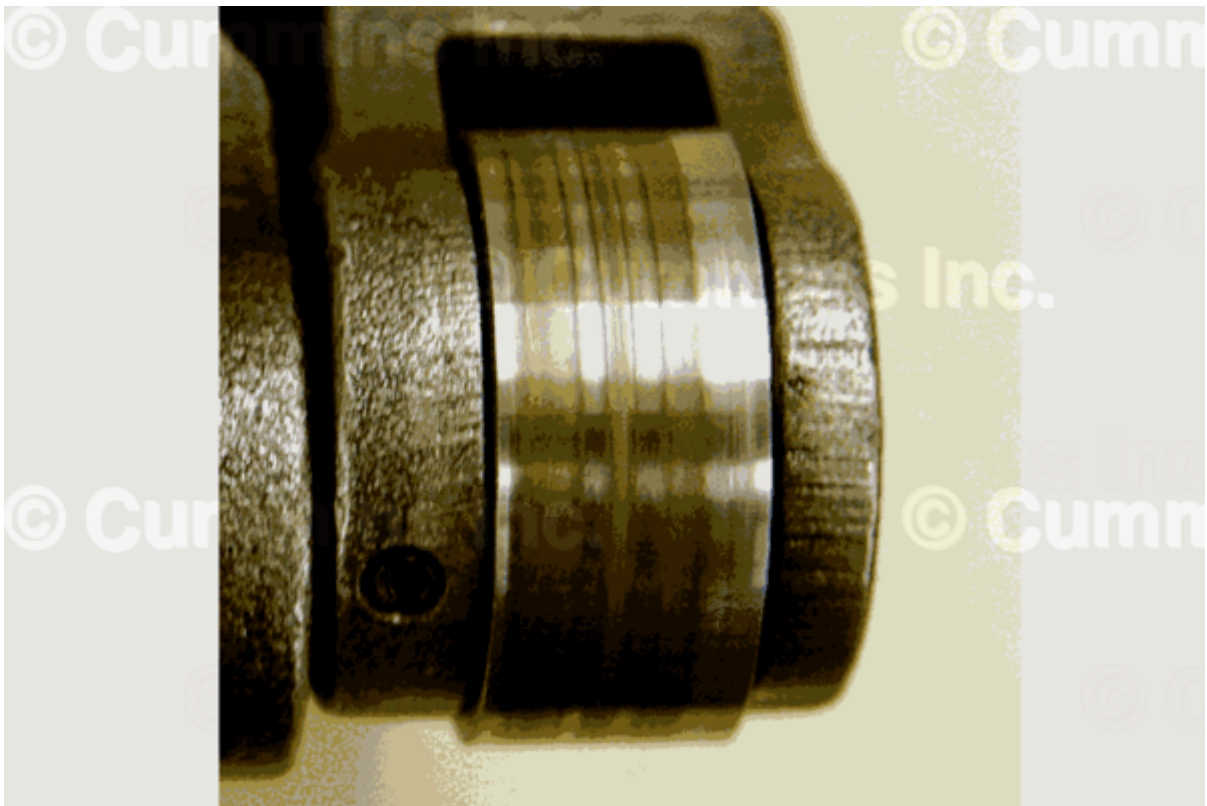


C. Denting: Unconditional Reuse

©Cummins Inc

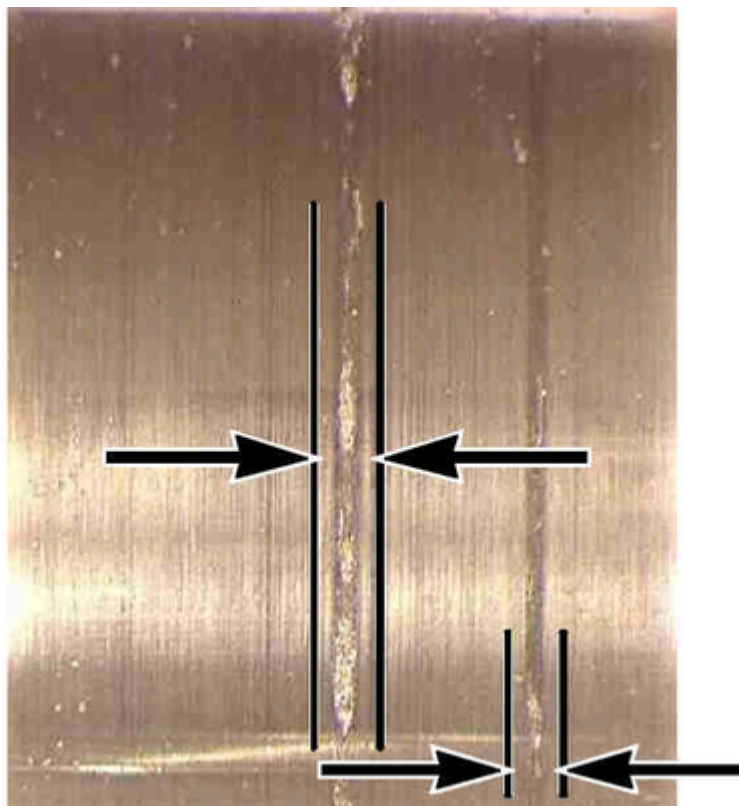


D. Frosted Bands: Unconditional Reuse

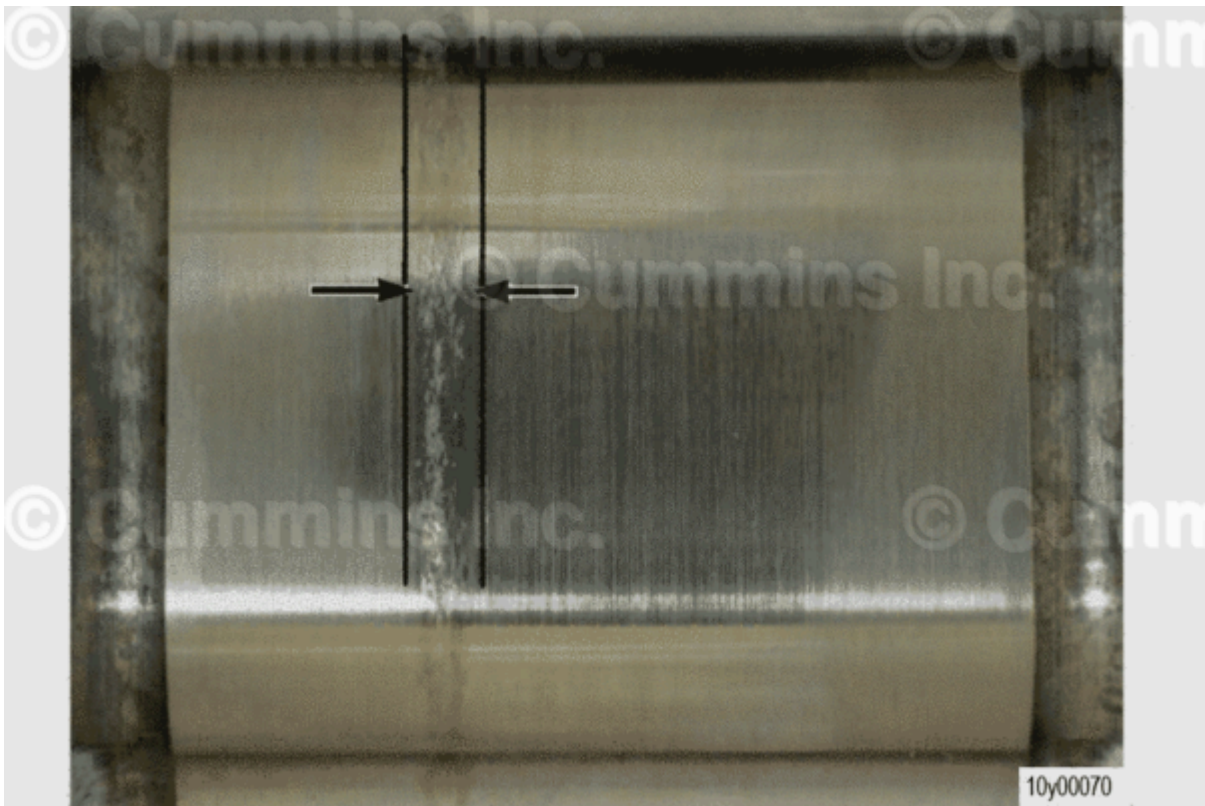


E. Varnished Bands: Unconditional Reuse

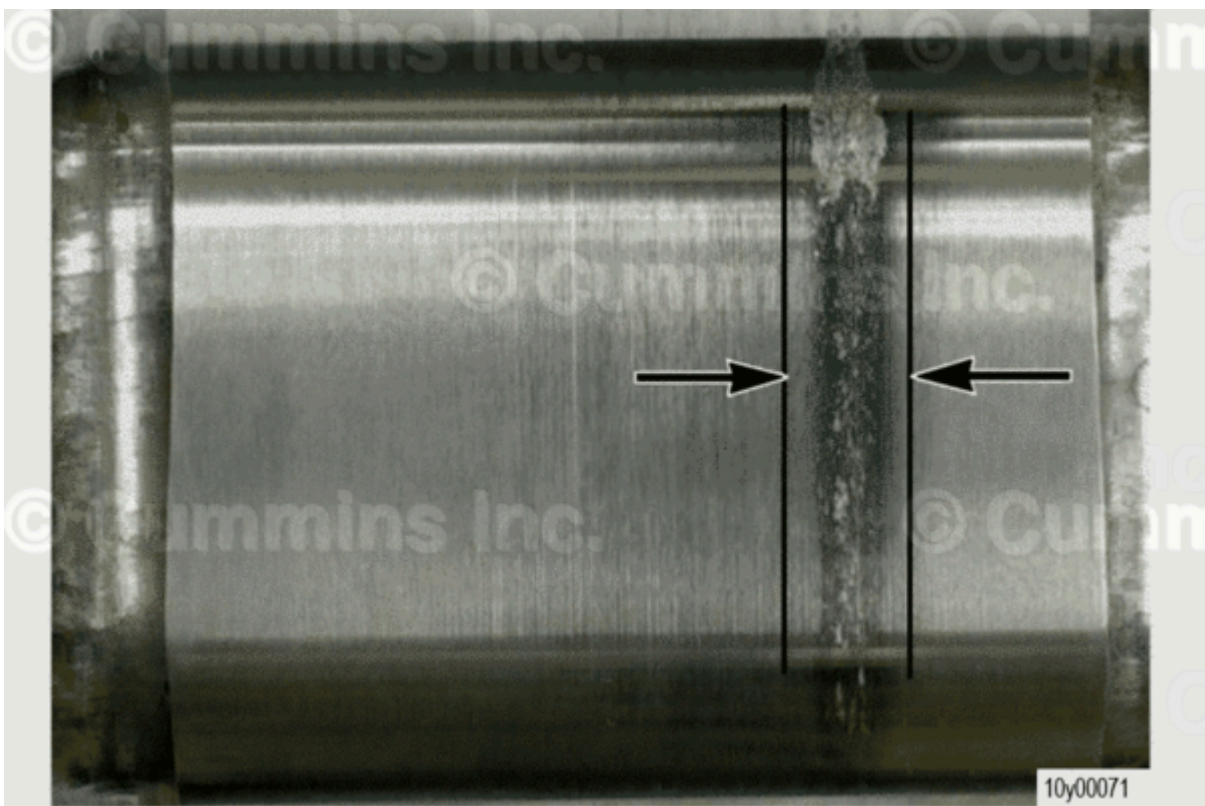
©Cummins Inc



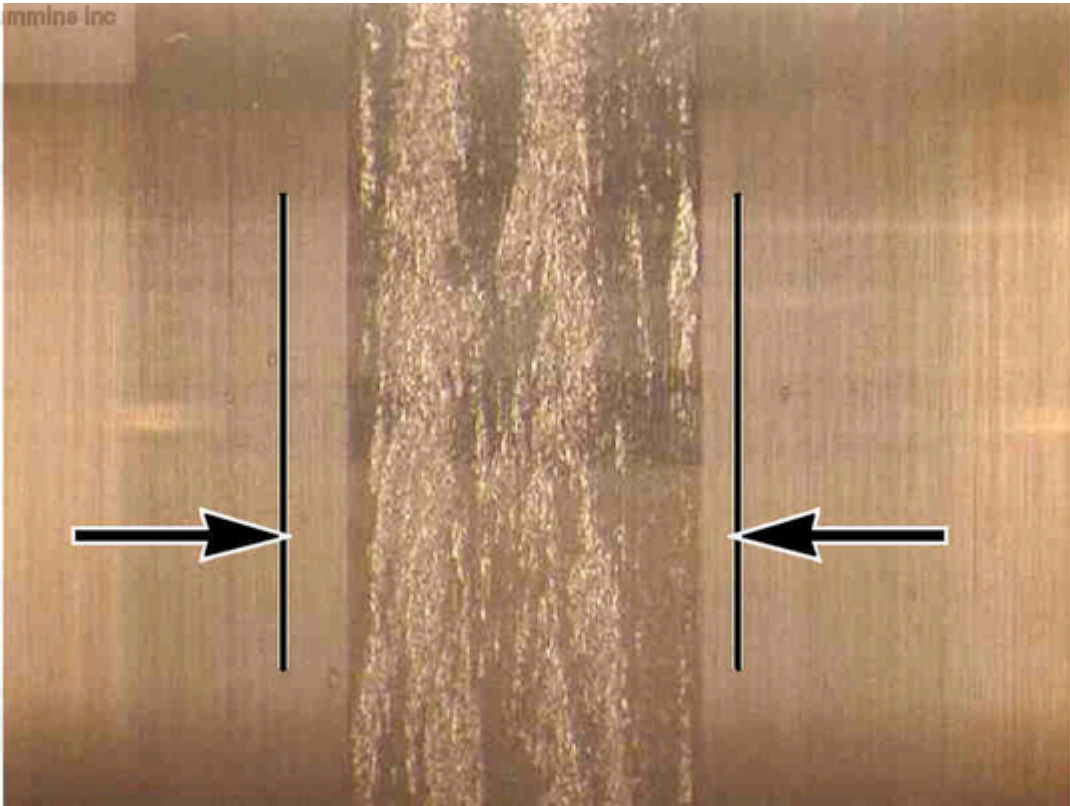
F. Galling Surface Distress: Conditional Reuse



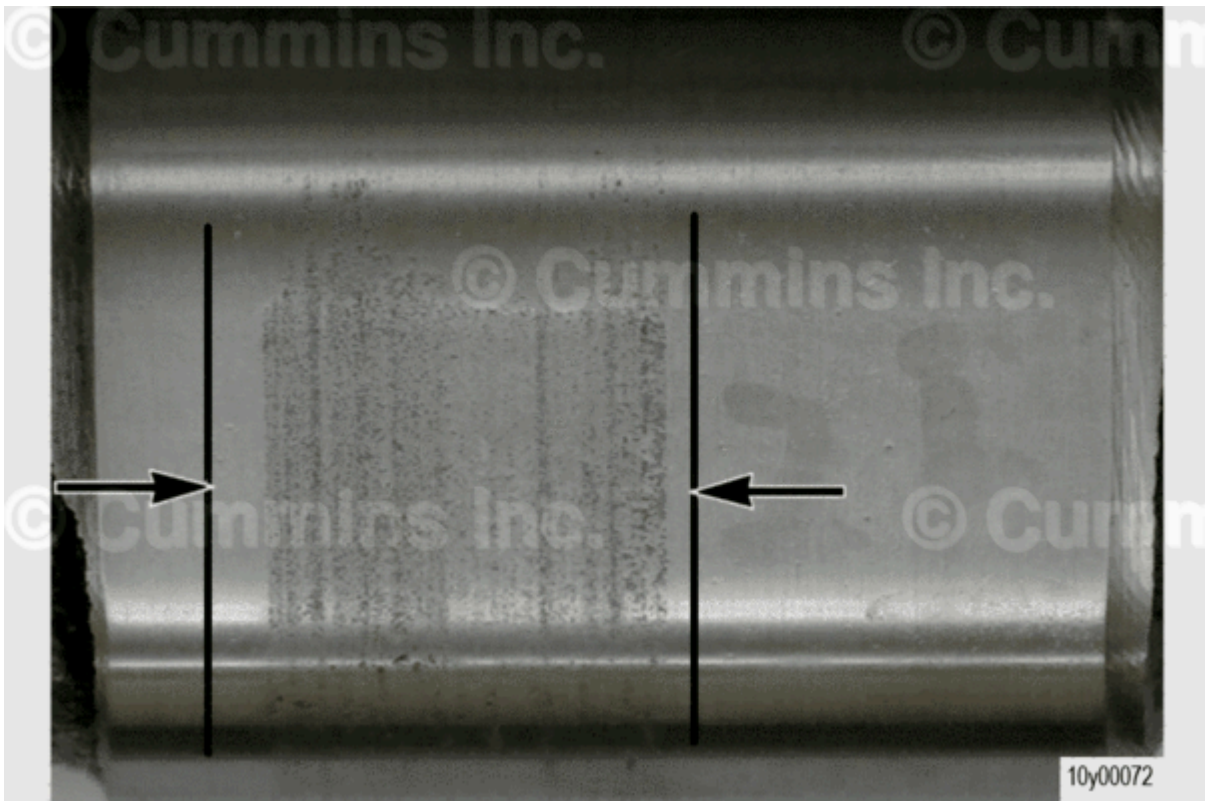
G. Galling Surface Distress: Conditional Reuse



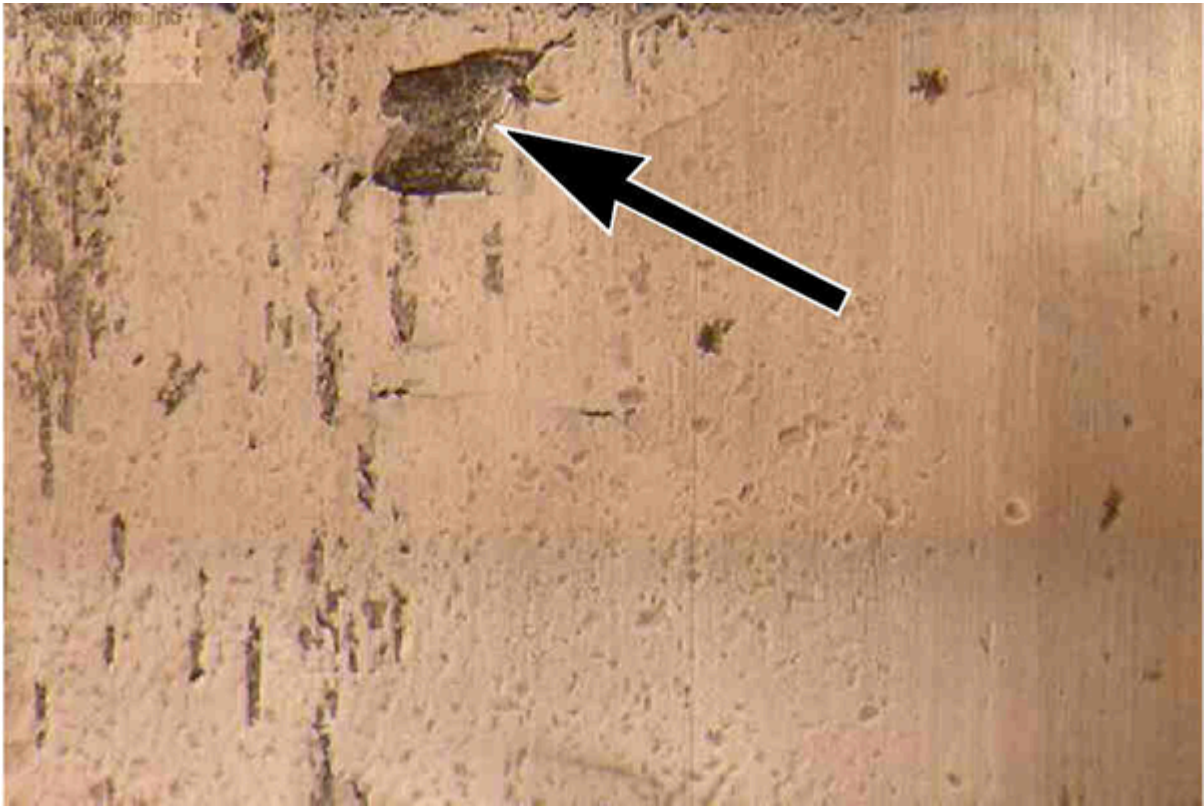
H. Galling Surface Distress: Conditional Reuse



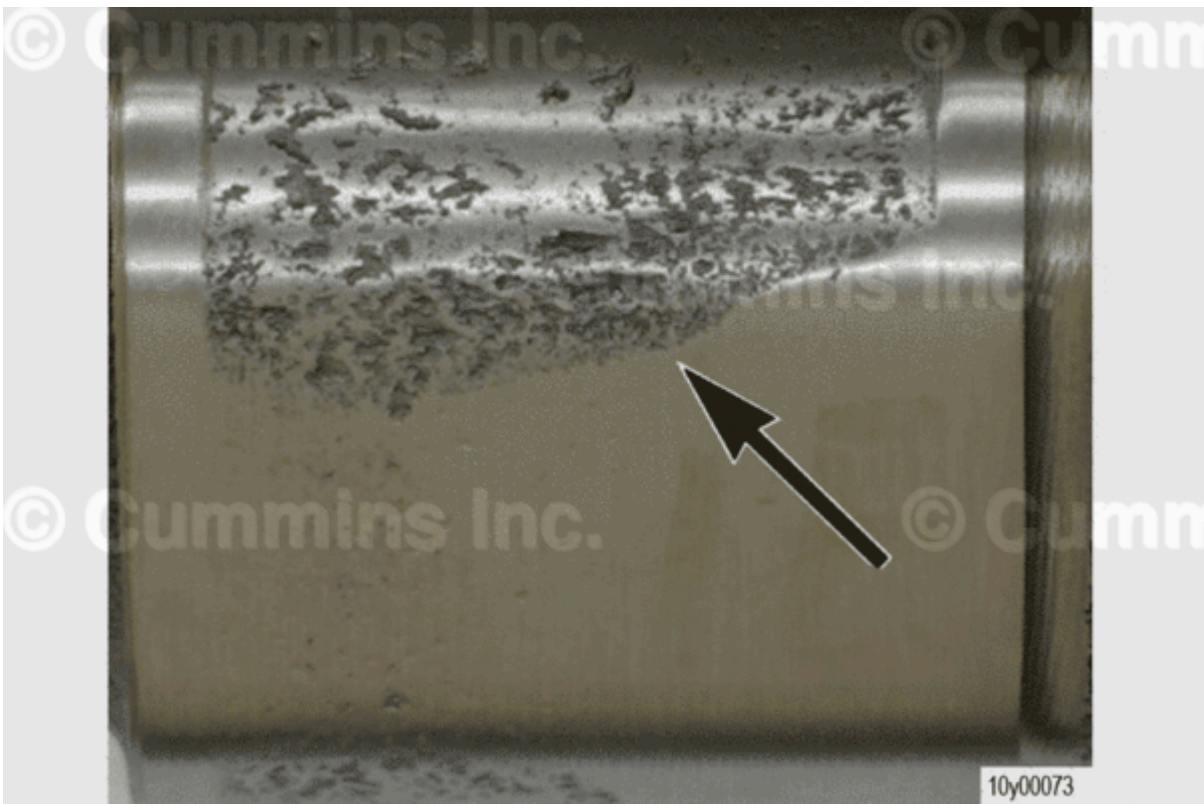
I. Galling Surface Distress: Do Not Reuse



J. Galling Surface Distress: Do Not Reuse



K. Pitting: Do Not Reuse

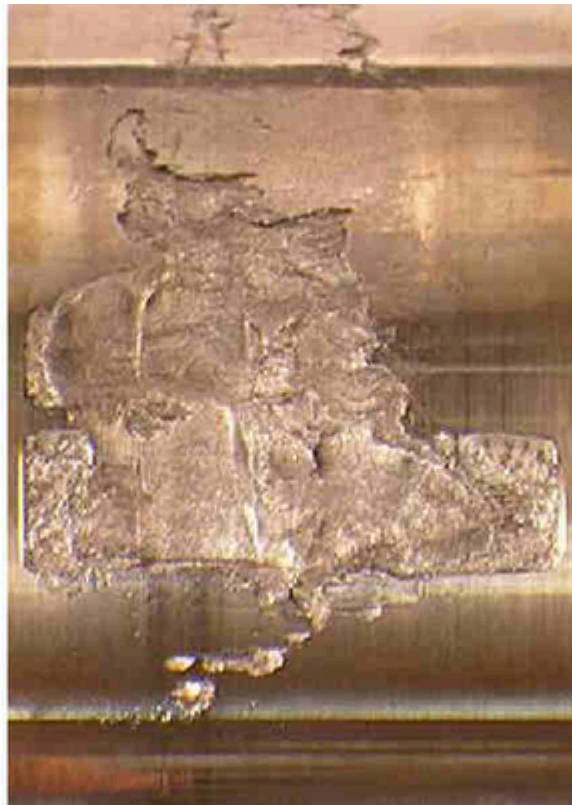


L. Pitting: Do Not Reuse



M. Pitting: Do Not Reuse

©Cummins Inc



N. Macrospalling: Do Not Reuse

Table 1: Unconditional Reuse Guidelines for Figures A, B, C, and D

Figure	Condition of Lobe	Reuse	Engines Affected
A	New	Unconditional reuse	All
B	Polish, no damage	Unconditional reuse	All

C	Denting	Unconditional reuse	All
D	Frosted bands	Unconditional reuse	All
E	Varnished bands	Unconditional reuse	All
F	Galling ¹ - Less than 1.5 mm [1/16 in] wide surface distress, no pits	Conditional reuse ²	All other engines / camshafts not listed separately (including 15L engine valve camshaft)
	Galling ¹ - Less than 3 mm [2/16 in] wide surface distress, no pits	Conditional reuse ²	X12 camshaft only
	Galling ¹ - Less than 5 mm [3/16 in] wide surface distress, no pits	Conditional reuse ³	ISX Signature, QSX15 and ISX 15L with CM870 Control Module - injector camshaft only
G	Galling ¹ - More than 1.5 mm [1/16 in] wide surface distress, no pits	Replace	All other engines/camshafts not listed separately (including 15L engine valve camshaft)
	Galling ¹ - More than 3 mm [2/16 in] wide surface distress, no pits	Replace	X12 camshaft only
	Galling ¹ - More than 5 mm [3/16 in] wide surface distress, no pits	Replace	ISX Signature, QSX15 and ISX 15L with CM870 Control Module - injector camshaft only
H	Pitting	Replace	All
I	Macrospalling	Replace	All

Explanation of Notes:

1. If multiple gall streaks are present but **not** connected, apply width criteria to each streak individually.
2. Do **not** reuse camshaft if engine is being overhauled or if camshaft is out of engine.
3. Do **not** reuse camshaft if engine is being overhauled.

Document History

Date	Details
24-jun-2015	Converting service bulletin file type
2017-12-19	Changed Explanation of Notes to numbers from bullet points.
2019-11-6	Adding additional pictures and X12 specific data
2020-6-8	Corrected typo.
2022-8-8	Changed the wording in the Lubricating Oil System Inspection section.
2025-6-30	Removed from service bulletin that frosted bands are not detectable with a fingernail from service bulletin content.

Last Modified: 30-Jun-2025
